

DocStock – Lab 1 Outline

Team Pearl · CS 410

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1. Introduction

Introduce the CS 410 product and the approach to demonstrating its characteristics through prototyping (essentially an abstract).

- 1.1. Summarize the societal problem with some reference/date support.
- 1.2. Outline the problem and needed solution characteristics.
- 1.3. Briefly introduce your product by name.

2. DocStock Product Description

Provide a top-level description of the CS 410 product for the average reader. Provide a summary of the solution – and its goals and objectives. This section should be one paragraph minimum.

2.1. Key Product Features and Capabilities

What does it do? What is significant / unique / innovative about it? What does it accomplish? Describe how this solves the problem.

2.2. Major Components (Hardware/Software)

Provide an overview of the hardware needed to support the solution. Describe how it is structured based on CS 410 MFCD. Define and describe the software to be developed.

3. Identification of Case Study

For whom is this product being developed? Why? Who else might use this in the future?

4. Glossary

Alphabetical list of terms and abbreviations.

Adherence – the extent to which a patient takes medication as prescribed (correct timing, dosage, frequency, and duration).

Backend – the server-side component housing DocStock's business logic and database (users, medications, schedules, history, supply data).

Caregiver access – a feature letting a designated person view or manage another user's medication profile.

CI/CD – Continuous Integration / Continuous Deployment; the pipeline for automated testing and deployment.

Dashboard – the centralized web interface where a user views and manages all of their prescriptions.

Dose – a single scheduled administration of a medication.

Drug interaction – an effect occurring when two or more medications affect one another; DocStock screens each new medication against the user's active list.

Git / GitHub – the version-control system and hosting platform used to develop DocStock.

IDE – Integrated Development Environment; the code editor used for the project (Visual Studio Code).

MFCD – Major Functional Component Diagram (confirm this expansion against the CS 410 course materials before submitting).

Nonadherence – failure to take a medication as prescribed; a driver of missed, delayed, or duplicated doses.

Prescription – a medication ordered for a patient by a licensed provider.

Refill alert – a notification sent before a medication's supply runs out.

Regimen (medication schedule) – the plan of which medications to take and when.

Reminder – a notification sent when a scheduled dose is due.

Supply monitoring – tracking of how much of each medication remains.

Web-based application – software accessed through a browser rather than a native mobile app.

5. References

Listing of sources (IEEE).

1. Centers for Disease Control and Prevention, "CDC Grand Rounds: Improving medication adherence for chronic disease management – Innovations and opportunities," MMWR, vol. 66, no. 45, pp. 1248–1251, Nov. 2017.
2. Medisafe, "App Features," medisafeapp.com.
3. MyTherapy, "MyTherapy: Medication Reminder & Pill Tracker," mytherapyapp.com.